Power Wave 4

Three Phase, 8kW to 400 kW

A POWERFUL INVERTER SOLUTION

Best high-capacity Three Phase solution you can get, UL1778 and UL924 listed and meeting strict standards from NFPA — with its versatile design and exceptional reliability, the Power Wave 4 inverter is the ultimate solution to meet all your emergency lighting needs.

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For the latest product news, technical specs, drawings, guides and manuals.

STANDARD FEATURES

- 90 Minute Backup Design at Full Load
- Listed to UL924 and UL1778 Standards
- Compatible with Self-Ballast Fluorescent, Incandescent, Halogen, Quartz Re-strike, HID, HPS and LED
- True Online Double Conversion Technology with **Zero Transfer Time**
- Pure Sinusoidal Waveform
- High Frequency Pulse Width Modulation (PWM) for Conditioned Power to the Load
- Internal Maintenance Bypass Switch
- Field Selectable: UPS Double Conversion or Green Mode (Standby)
- Independent Phase Control
- Global Monitoring (GMS 1): Local Event Logger with Memory (PC Connections vis RS232 and RS485) 500 Events
- Input Breaker, Output Breaker and DC Breaker
- Microprocessor Controlled LCD with visual status and Event indicators
- Self-Testing and Battery Exerciser with LCD Display

OPTIONAL FEATURES

- Global Monitoring (GMS):
  - GMS 2: Remote Monitoring with Local Network Connection (WiFi, Ethernet or Modem)
  - GMS 3: Remote Monitoring with Cellular Network Connection
- Remote Status Panel
- Battery Monitoring (String or Per Battery)
- TVSS Input Transient Voltage Surge Suppressor (Class C)
- Form C Dry Contact N/O (standard) or N/C (available at the time of order)
- Normally ON or Normally OFF Auxiliary Output Circuit Breakers (20 Max)
- Custom Rated KAIC Breakers
- 65 KAIC Total System Short Circuit Current Rating
- Thermal Runaway Control (IFC 1206.2)
- External Maintenance Bypass Switch
- Seismic Brackets
- Long Life Battery 20 Year Warranty
- Maintenance:
  - Extended Warranty
  - Preventative Maintenance Plans

ISO9001 Certified Since 1996

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120-DS-11-23

NFPA CODES & STANDARDS COMPLIANT

MADE IN USA

SPECIFICATIONS

Power Wave 4

Three Phase
8 to 400 kW

*All units are 90 minutes battery backup time @ full load. Consult factory for backup times (up to 120 minutes)

SAFETY STANDARDS
UL924, UL1778, NFPA101, NFPA70, NEMA, and ANSI
Listed to UL924 and UL1778 Standards

INPUT
AC Voltage: ±15%
Frequency: 50/60Hz ±7

OUTPUT
AC Voltage Regulation: ±1%
Wave Form: Sinusoidal
Harmonic Distortion: Less than 2% THD
Frequency Tolerance: ±0.1 Hz

PROTECTION
Overload: Less than 110% Load: Continuous, 125-150%: 5 min, Higher than 150%: 30 sec

BATTERY
Battery: Sealed, Maintenance-free, Lead-Acid, VRLA (Standard) 10 years
Recharge Time: Varies per kVA, conforms to UL924

NOISE ISOLATION
Isolation: True Galvanic Isolated

ENVIRONMENT (Electronics)
Operating Temperature: 0° to 40°C (32° - 104°F)
Storage Temperature: -20° to 70°C (-4° - 158°F)
Humidity: 0% to 90% (non-condensation)
Altitude: Up to 5,000 ft
Audible Noise: Less than 65dBA at 1 meter/39.4 in.

Cabinet Dimensions

(Inches) Width x Height x Depth

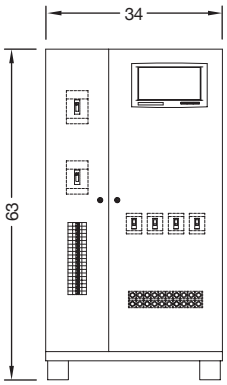
Cabinet I
34"W x 63"H x 31.5"D

Cabinet II
55.5"W x 63"H x 31.5"D

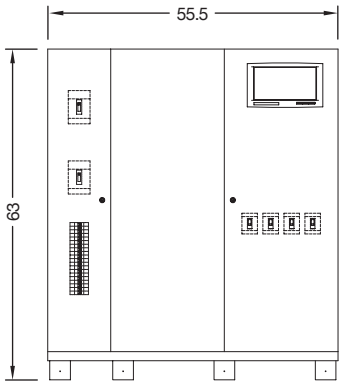
Battery Cabinet
51"W x 70"H x 30.5"D

Each model comes with both inverter and battery cabinets. Larger capacity will require additional battery cabinets. Size and weight will vary depending on capacity

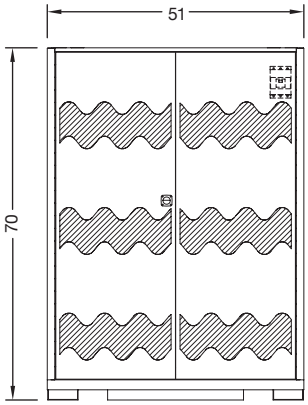
Cabinet I
10 - 60kVA



Cabinet II
80 - 160kVA



Battery Cabinet



Model Numbers

Unit Name	Capacity kVA/kW		Input (Volts)		Output (Volts)	
PW	**010**	10/8	**B**	208Y/120	**05ATT3**	208Y/120
Power Wave 4	**020**	20/16	**H**	480Y/277	**09ATT3**	480Y/277
	030	30/24				
	040	40/32				
	050	50/40				
	060	60/48				
	080	80/64				
	100	100/80				
	120	120/96				
	160	160/128				
	200	200/160				
	240	240/192				
	300	300/240				
	400	400/320				
	500	500/400				

kW equates to real power.

BTU / Weight

Capacity kVA/kW	BTU/HR Standard	BTU/HR Green	Cabinets - 90min (lbs) (# of Cabinets) Inverter	Top	Front	Cabinets - 120min (lbs) (# of Cabinets) Inverter	Top	Front
10/8	3374	1742	840 (1)	2430 (1)	2975 (1)	840 (1)	2430 (1)	2975 (1)
20/16	6747	2873	1083 (1)	3213 (1)	3990 (1)	1083 (1)	3416 (1)	3990 (1)
30/24	10120	4310	1260 (1)	4361 (1)	4135 (1)	1260 (1)	4535 (1)	5034 (1)
40/32	12131	5747	1414 (1)	3213 (2)	5034 (1)	1414 (1)	3416 (2)	3990 (2)
50/40	15164	5687	1525 (1)	3427 (2)	3990 (2)	1525 (1)	3867 (2)	4153 (2)
60/48	18197	6824	1724 (1)	3867 (2)	4135 (2)	1724 (1)	4923 (2)	5034 (2)
80/64	24263	9099	2276 (1)	4615 (2)	5034 (2)	2276 (1)	3707 (1) 4729 (2)	5034 (3)
100/80	30329	11373	2984 (1)	3707 (1) 4729 (2)	5034 (3)	2984 (1)	3416 (2) 4923 (2)	5034 (3)
120/96	32395	10130	3138 (1)	3416 (2) 4923 (2)	5034 (3)	3138 (1)	4923 (4)	5034 (4)
160/128	43193	13507	3868 (1)	*	5060 (4)	3868 (1)	*	5060 (5)
200/160	53992	16884	5746 (1)	*	5060 (5)	5746 (1)	*	5060 (6)
240/192	64790	20206	6229 (1)	*	5060 (6)	6229 (1)	*	*
300/240	80988	25326	7293 (1)	*	*	7293 (1)	*	*
400/320	94942	33768	9061 (1)	*	*	9061 (1)	*	*
500/400	118678	42210	10166 (1)	*	*	10166 (1)	*	*

The approximation is worst case BTU output, measured during recharge following a discharge.

Warranty

Standard warranty is eighteen (18) months from ship date or twelve (12) months from start up, whichever occurs first. Optional second year warranty with factory performed preventative maintenance available.

ISO9001 Certified Since 1996

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MADE IN USA

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ORDERING GUIDE

PROJECT NAME :

REP / DISTRIBUTOR :

ORDER NUMBER **DOES NOT** REFLECT MODEL NUMBER. PLEASE CONSULT FACTORY.

Power Wave 4

Example: PW-048-B05-S90-N08A40-FT-FC-GMS3-PM2

ELECTRICAL DESIGNER / ENGINEER :

DATE :

PW-

PW — MODEL SERIES

PW: Power Wave 4

048 — CAPACITY (KW)

08:	8kW
16:	16kW
24:	24kW
32:	32kW
40:	40kW
48:	48kW
64:	64kW
80:	80kW
96:	96kW
128:	128kW
160:	160kW
192:	192kW
240:	240kW
320:	320kW
400:	400kW

B05 — VOLTAGE INPUT/OUTPUT

INPUT	OUTPUT
B - 208Y/120	**05** - 208Y/120
H - 480Y/277	**09** - 480Y/277

*Input 'B' not compatible with Output '09'

S90 — BATTERY TYPE

S90: Standard @ 90min
* **L90:** Long Life @ 90 min
H90: Hi Temp @ 90 min
U90: USA @ 90 min
S120: Standard @ 120min
* **L120:** Long Life @ 120 min
H120: Hi Temp @ 120 min
U120: USA @ 120 min

*Long Life 20 Yr. Pro Rata Battery.

N08A40 — AUX BREAKERS (UL1077)

AUX BREAKERS EXAMPLE

N 08 A 40

OUTPUT

N: Normally ON
F: Normally OFF
****FD:** Normally OFF w/ Time Delay

BREAKERS

***01 - 20** (Max 20)

VOLTAGE

A - 120V, 1P
B - 208V, 2P
C - 240V, 2P
D - 277V, 1P
E - 480V, 2P
F - 208Y/120, 3P
G - 480Y/277, 3P

AMP RATING***

10, 15, 20, 25, 30, 40, 50, 60
STANDARD: 20 AMP

* Only up to 20 poles.
** Allows you to have individual circuits remain on battery even when utility is restored for a period of time. Must be a normally OFF circuit.
***Consult factory for custom amp rating.

FT + FC + GMS3 — OPTIONS

PICK ANY:
X: Seismic Mounting
LCR20A(#): Load Control Relay, Single Flex Stud Mounting (#): Qty
LCR20(#): Load Control Relay, No Flex (#): Qty
EB: Ex. Maintenance Wraparound Bypass Switch
TVSS: Input Transient Voltage Surge Suppressor (Class C)
SCCR: 65 KAIC Total System Short Circuit Current Rating
DS: Drip Shield Hood
CL: Corbin Lock

BATTERY OPTIONS
FC: Fast Charge (12 hrs or less)
TR: Thermal Runaway Control (IFC 1206.2)

MONITORING OPTIONS
DCT2: Dual, Open and Closed Dry Contacts
A: Local Audible Alarm with Silencer Switch
CP: RS 232 Connecting Port
P: Remote Status Panel (Hard Wired with Extended Cable)

SYSTEM TRANSFER
TD(B): User Selectable Time Delay to Battery for Capacitor Discharge During EM (Standard 45 Millisecondst)
TD(U): User Selectable Time Delay Back to Utility During Power Restoration for Facility Inrush

BATTERY MONITORING SYSTEM (BMS) PICK ONE:
SM: String Monitoring
BM: Battery Monitoring
BM-TS: Temperature Sensor

GLOBAL MONITERING SYSTEM (GMS) PICK ONE:
GMS 2: Testing Logs with Remote Accessibility Through Local Network
GMS 3: Testing Logs with Remote Accessibility through Local Network and GPRS for Cellular Text Alerts

PM2 — WARRANTY & SERVICES

PICK ANY:
Blank: Standard Onsite Startup and Warranty

ADDITIONAL:
EXT: Additional One Year Extended Warranty (Includes First Year Preventative Maintenance)
PM (2-3): Onsite Preventative Maintenance (Once Per Year, up to 3 Years)
T1: 4 Hour Training at Time of Start Up
T2: Additional Training day up to 8 hours

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